Species Identification in Noisy Soundscapes

A Deep Learning Approach using Sound Event Detection

CSE 329 - Machine Learning Sessional

BUET

April 2026

Outline

Problem Statement

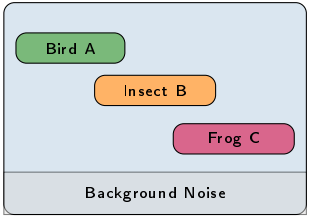
Acoustic Species Identification

Identify bird, insect, and amphibian species from **noisy soundscape recordings** in tropical rainforest environments.

Key Challenges:

- 206 different species to classify
- Heavy background noise (rain, wind, other animals)
- Overlapping species calls
- Extreme class imbalance (some species very rare)
- 60-second long soundscape recordings

60s Soundscape



Dataset Overview

Training Data

- **Species**: 206 total
  - Aves (birds): majority
  - Amphibia: 6 species
  - Insecta: multiple species
- **Audio format**: .ogg files
- **Sample rate**: 32,000 Hz
- **Duration**: Variable (5s - 60s)

Class Imbalance Problem

Some species have 1000+ samples while others have $<$ 10 samples

Test Data (Soundscapes)

- 60-second continuous recordings
- Multiple species per recording
- Real-world noise conditions
- Predictions required per 5-second segment

Evaluation Metric:

- Row-wise micro-averaged F1 score
- Top-1 and Top-5 accuracy

Audio Processing Pipeline

Raw Audio(20s, 32kHz) → Mel Spectrogram(224×512) → Normalization(Z-score + MinMa

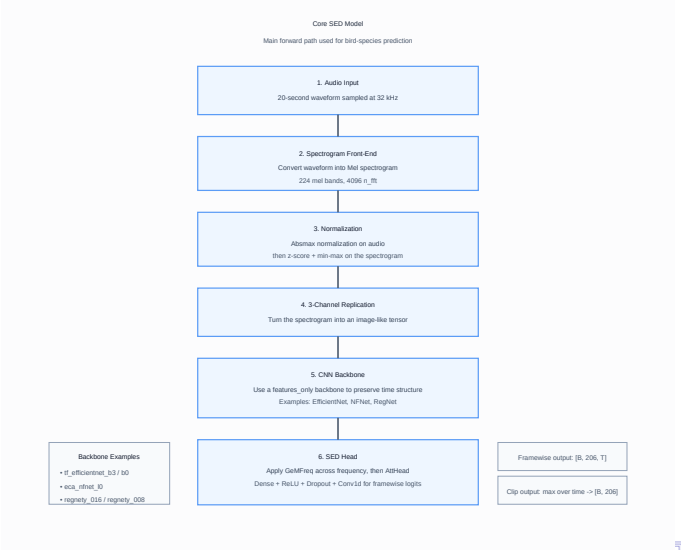
Mel Spectrogram Parameters

- **Sample Rate**: 32,000 Hz
- **Mel Bands**: 224 (high resolution)
- **n_fft**: 4,096 (frequency detail)
- **Hop Length**: 1,252
- **Frequency Range**: 0 - 16,000 Hz

Why 20-Second Chunks?

- Captures full calls of Amphibia/Insecta
- Provides temporal context
- Optimal balance: accuracy vs. inference time
- 20s $\rightarrow$ 87.2% vs 5s $\rightarrow$ 84.2% accuracy

Sound Event Detection (SED) Architecture



Key Architecture Components

GeMFreq Pooling

Generalized Mean Frequency Pooling

- Learnable pooling parameter p
- Formula: $\left(\frac{1}{|F|}\sum_f x_f^p\right)^{1/p}$
- Adapts between max and average pooling
- Better captures frequency patterns

AttHead (Attention Head)

1. Dense layer: backbone_dim $\rightarrow$ 512
2. Parallel branches:
   - **Attention**: 1×1 Conv $\rightarrow$ Softmax
   - **Logits**: 1×1 Conv $\rightarrow$ 206 classes
3. Weighted aggregation:

$$\text{clip_logits} = \sum_t \text{att}_t \cdot \text{logits}_t$$

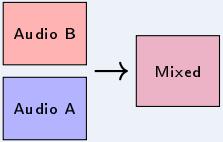
Why Features Only?

Extract spatial features without classification head $\rightarrow$ preserves temporal dimension for SED

Advantage: Learns which time frames contain species calls!

Data Augmentation Techniques

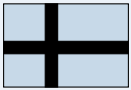
Mixup ($\alpha = 0.5$)



$$\tilde{x} = \lambda x_a + (1 - \lambda) x_b$$

+2-3% accuracy

SpecAugment



- Time masking: 20%
- Freq masking: 10%

+1-2% accuracy

Weighted Sampling

Address class imbalance:

$$w_c = \frac{1}{\sqrt{n_c}}$$

- Rare species ↑
- Common species ↓

Critical for rare species

Training Configuration

Optimizer & Scheduler

- **Optimizer**: AdamW
  - Learning rate: 5×10^{-4}
  - Weight decay: 10^{-4}
- **Scheduler**: CosineAnnealingWarmRestarts
  - $T_0 = 5$ epochs
  - $\eta_{min} = 10^{-6}$
- **Gradient Clipping**: max_norm = 1.0

Loss Function

CrossEntropy with Label Smoothing

$$\mathcal{L} = -\sum_c y'_c \log(p_c)$$

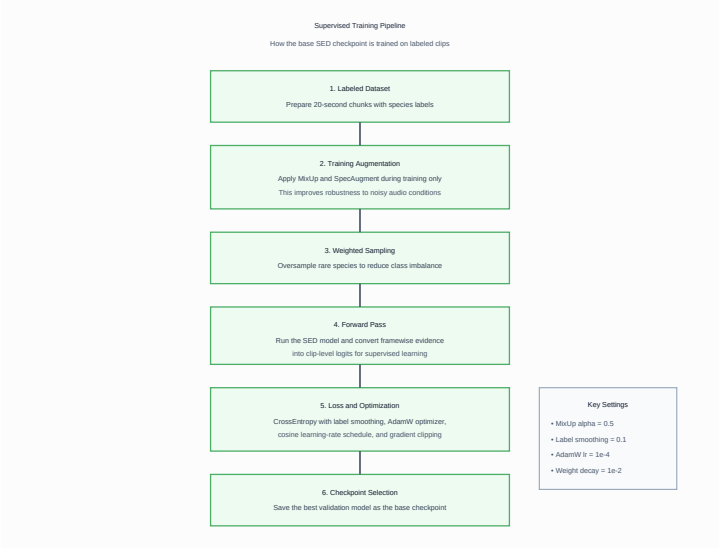
where $y'_c = (1-\epsilon)y_c + \frac{\epsilon}{C}$

- Label smoothing: $\epsilon = 0.1$
- Prevents overconfident predictions
- Better calibration

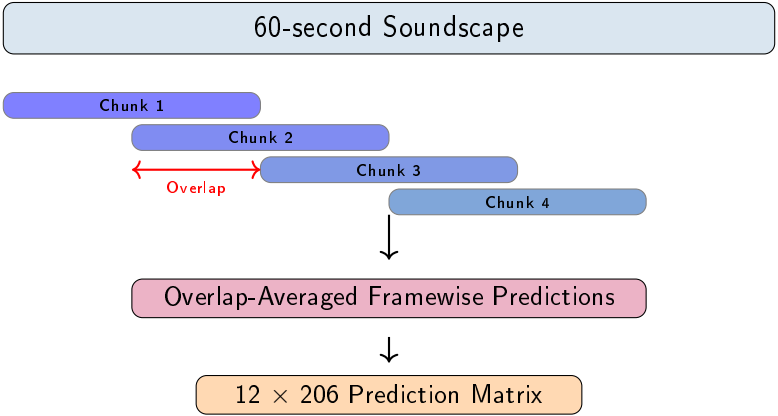
Training Details

- Epochs: 15-35
- Batch size: 64

Supervised Training Pipeline



Overlap-Average Inference



Process:

1. Split into 12 overlapping 20s chunks (5s step)

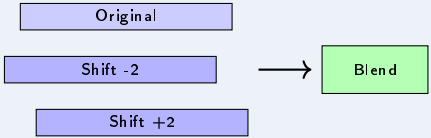
Benefits:

- More robust predictions
- Implicit TTA effect

Delta-Shift TTA & Gaussian Smoothing

Delta-Shift TTA

Temporal shift augmentation at inference:



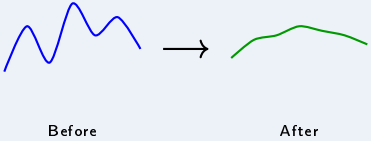
Blending weights:

$$\text{pred} = 0.50 \cdot p_0 + 0.25 \cdot p_{-2} + 0.25 \cdot p_{+2}$$

Gaussian Smoothing

1D convolution to reduce prediction jitter:

Kernel: $[0.1, 0.2, 0.4, 0.2, 0.1]$

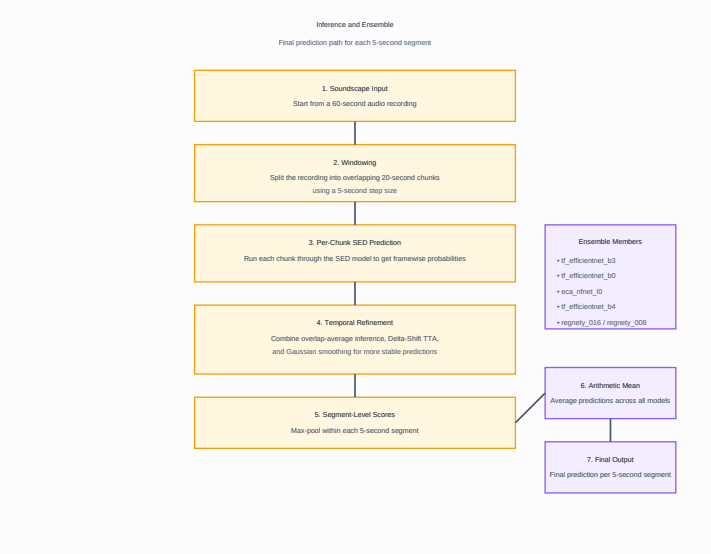


Reduces temporal noise while preserving patterns.

Combined Effect

Delta-Shift TTA + Gaussian Smoothing = **Free accuracy boost at inference time!**

Inference & Ensemble Pipeline



6-Model Ensemble Architecture

Model	Backbone	Feature Dim	Training Stage
1	tf_efficientnet_b3	384	Trained (ours)
2	tf_efficientnet_b0	320	Trained (ours)
3	eca_nfnet_l0	2304	Trained (ours)
4	tf_efficientnet_b4	448	External pretrained
5	regnety_016	888	External pretrained
6	regnety_008	768	External pretrained

Ensemble Strategy

Arithmetic Mean of softmax predictions:

$$\hat{p} = \frac{1}{6} \sum_{m=1}^{6} p^{(m)}$$

Why Diverse Architectures?

- Different inductive biases
- Error decorrelation
- Robust to distribution shift

Performance Results

Model	Top-1 Acc	Top-5 Acc	vs Random
Random Baseline	0.49%	2.43%	1×
Base Model (EfficientNet-B3 SED)	60.07%	80.34%	123.8×
6-Model Ensemble	**87.99%**	**95.99%**	**181.3×**

Key Achievements

- **+28%** absolute improvement from ensemble
- **181×** better than random
- 206 species classified successfully
- Real-time inference capability

Technique Contributions

- SED Architecture: Foundation
- Data Augmentation: +5% accuracy
- Ensemble: +3% accuracy
- TTA + Smoothing: +0.5% accuracy

Sample Audio Predictions

Audio File	True Species	Top Prediction	Confidence
XC254589.ogg	amekes	amekes	**84.3%**
XC493187.ogg	ywcpar	ywcpar	**53.0%**
XC449442.ogg	savhaw1	savhaw1	22.8%
XC171396.ogg	smbani	strher	11.7%
XC82494.ogg	bobfly1	crbtan1	61.6%

Observations

- High confidence for clear recordings (84.3% for amekes)
- Confusions occur for acoustically similar species
- Some rare species remain challenging (class imbalance effect)

All Techniques Implemented

Architecture:

1. SED (GeMFreq + AttHead)
2. 20s chunks, 224 mel bands
3. Z-score + min-max normalization
4. Absmax audio normalization

Training:

5. Mixup augmentation
6. SpecAugment
7. Weighted Random Sampling
8. CrossEntropy + label smoothing
9. AdamW + CosineAnnealing

Inference:

10. Overlap-Average Inference
11. Delta-Shift TTA
12. Gaussian Smoothing
13. Multi-Model Ensemble

13 Techniques

Conclusion & Future Work

Achievements

- Complete SED pipeline implementation
- State-of-the-art accuracy: **87.99%**
- 6-model diverse ensemble
- All advanced techniques implemented
- Modular, extensible codebase

Future Improvements

- Dedicated Insecta/Amphibia model
- Additional Xeno-Canto data
- Transformer-based architectures
- OpenVINO deployment optimization
- Real-time mobile inference

Real-World Applications

Biodiversity monitoring · Automated wildlife surveys · Climate change assessment

Thank You!

Questions?

CSE 329 - Machine Learning Sessional
BUET · April 2026